One-hot encoding: one column per category

Formation
A1 SH
A1 LM
B1 SH
A1 LM
B1 SH

→

A1 LM	A1 SH	B1 SH
0	1	0
1	0	0
0	0	1
1	0	0
0	0	1

Integer codes 0, 1, 2 would invent an order that linear models and KNN take literally

Median imputation plus a missing indicator

PE
3.1
NaN
2.8
NaN
4.0

→

PE	PE_missing
3.1	0
3.1	1
2.8	0
3.1	1
4.0	0

median of the training values (2.8, 3.1, 4.0) = 3.1

Both steps are fitted: categories and medians come from training rows only